CS & IT ENGINEERING



Algorithms

Dynamic Programming



Lecture No.- 01

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Min Cost Spanning Tree (mcst)

PYQs

Topics to be Covered



- Topic
- Topic
- Topic

Shortest Path Algos (Intro)

↳ 1) Dijkstra's Algo



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



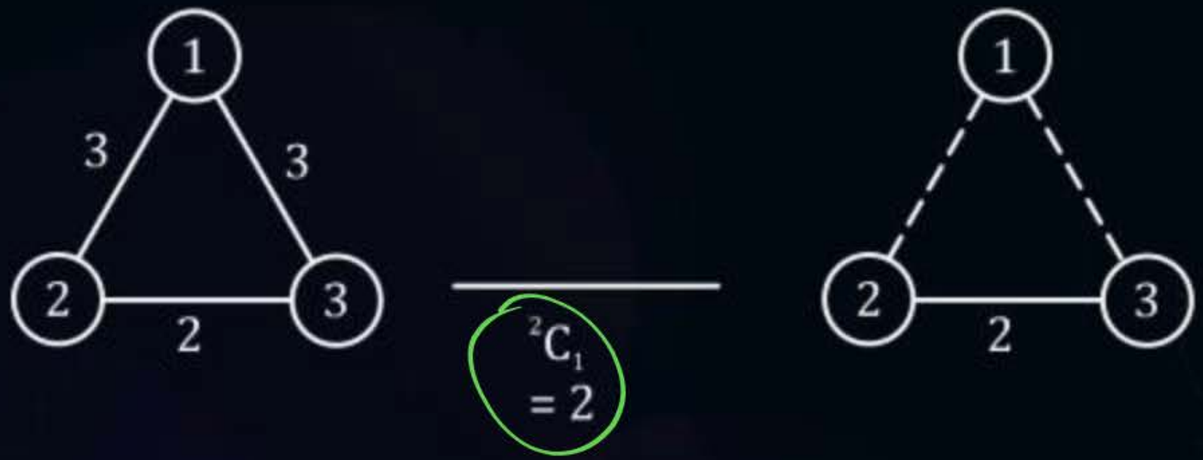
Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

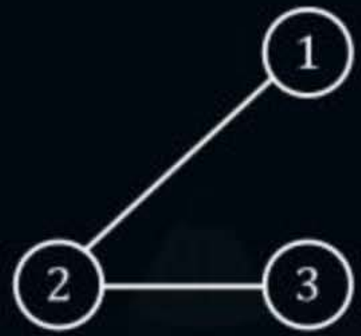


Topic : Greedy Techniques

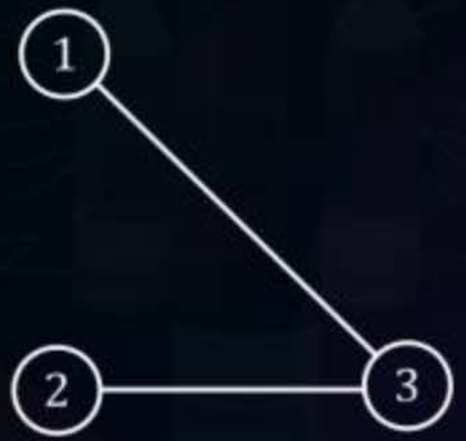
Concept:-

How many MCST?





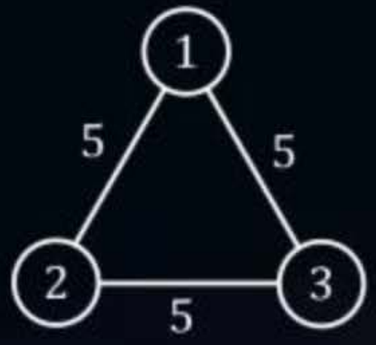
OR





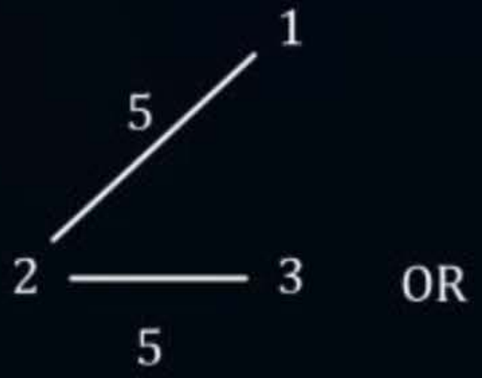
Topic : Greedy Techniques

Eg.1.

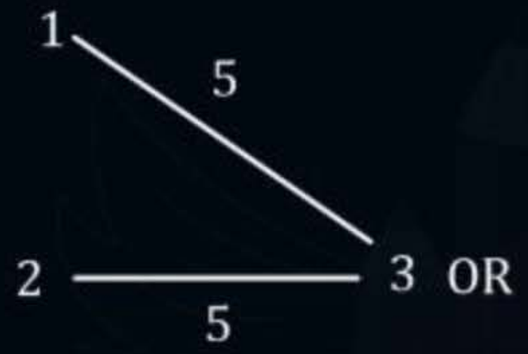


$^3C_2$

3. MCSTs



OR



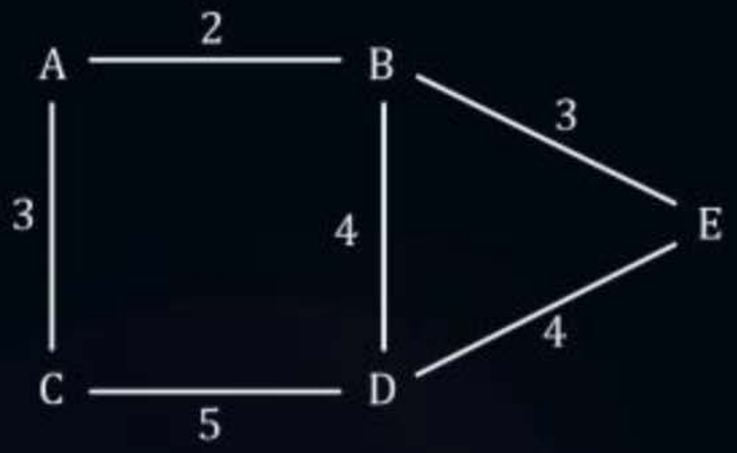
OR

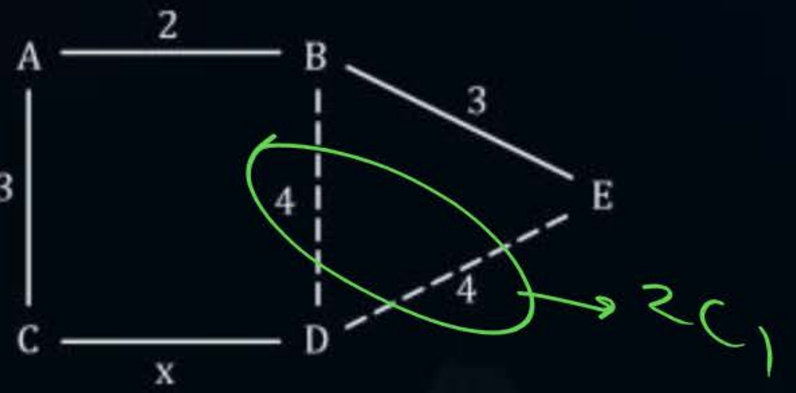


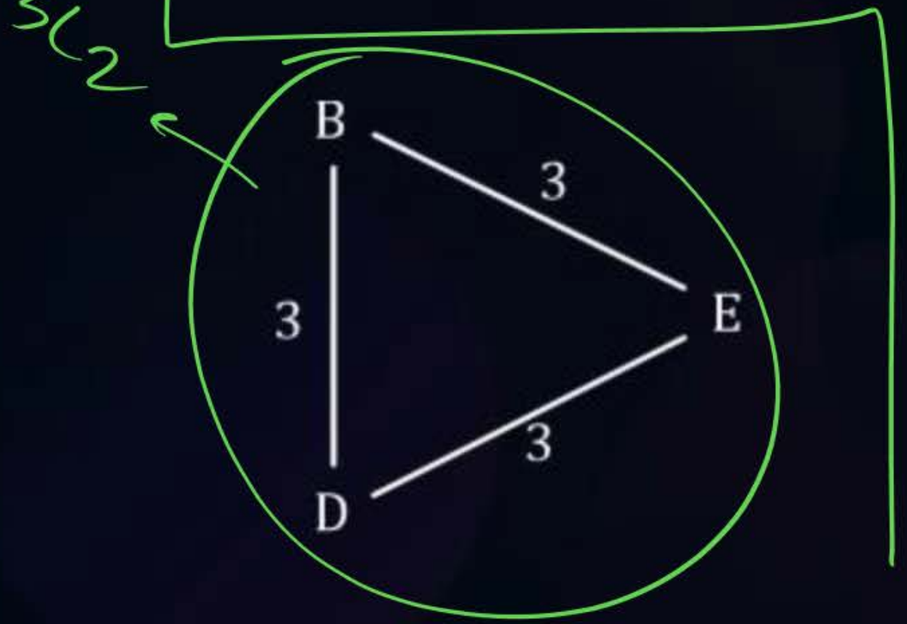


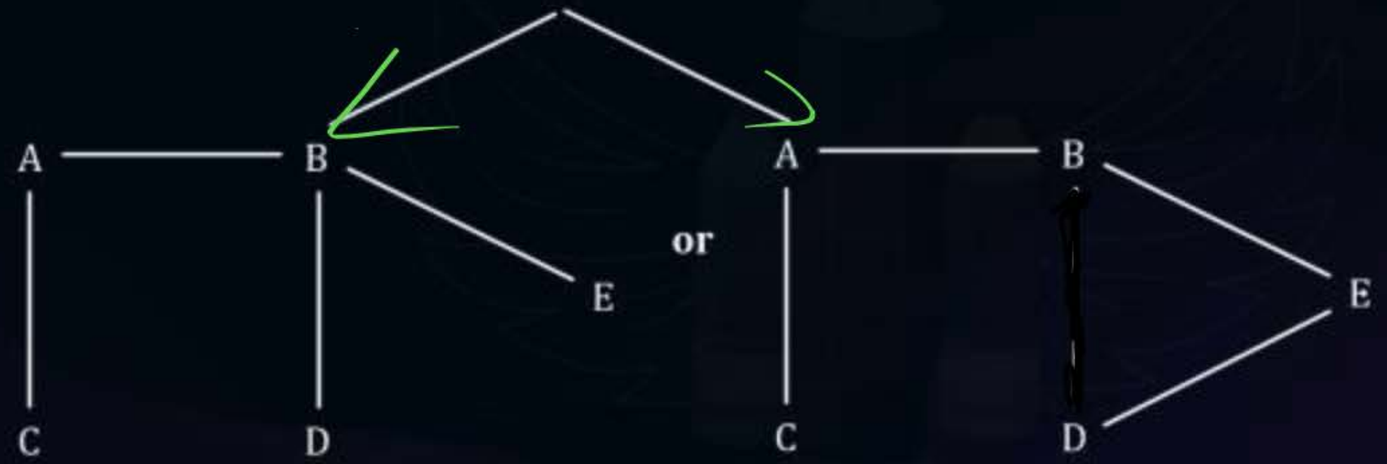
Topic : Greedy Techniques

Eg.2.









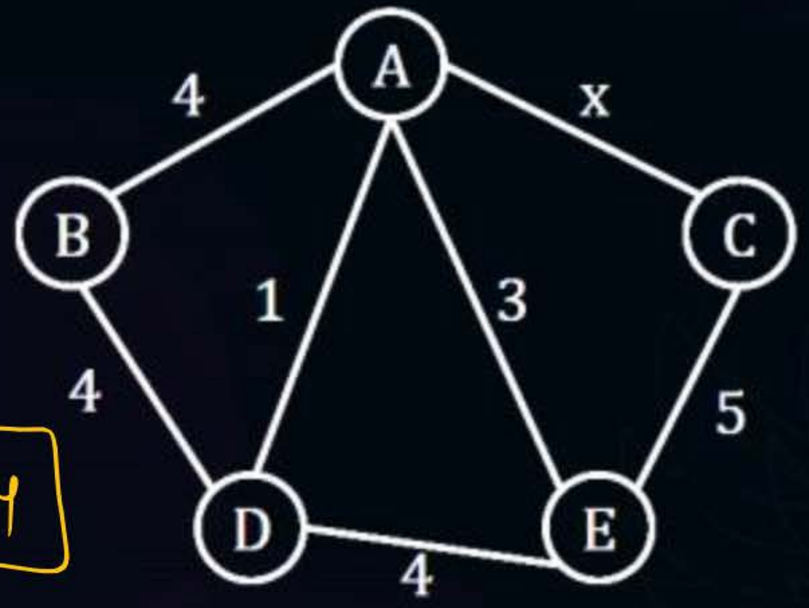
Topic : Greedy Techniques

45%

#Q. Consider the following undirected graph G:

PYQ

v. similar to Gate'25

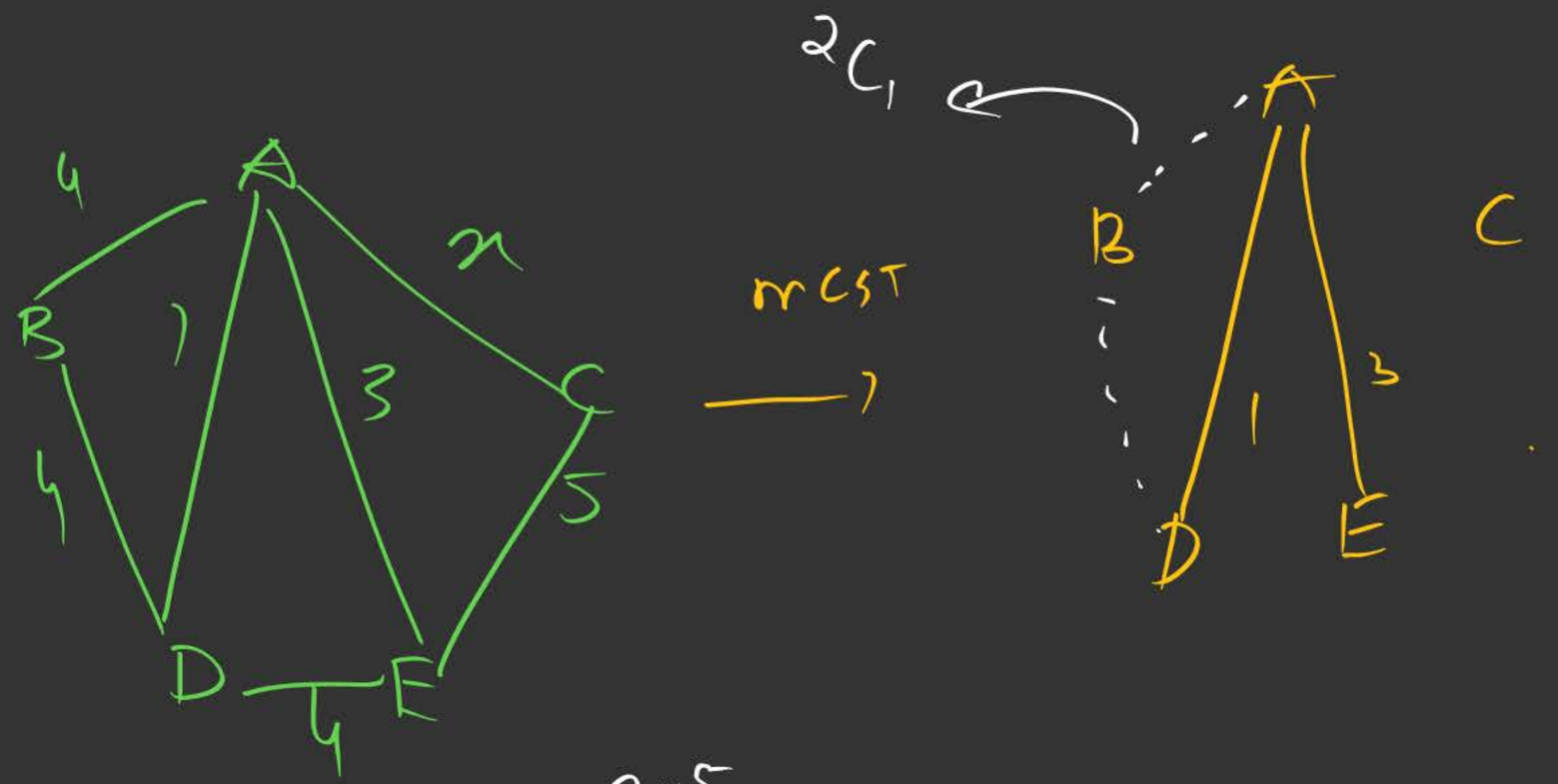


n=5

count=4

Choose a value for x that maximize the number of minimum weight spanning tree (MWSTs) of G. The number of MWST of G this value of x is_____.

for



$n = 5$

$e = n - 1$

$= 4$



Topic : Greedy Techniques

Possible Case:

1. $x > 5$: Then AC will never be a part go MCST and CE will be always a part.

   → Left → $2C_1$ Right →1 ⇒ Total = $^2C_1 * 1 = 2$ MCSTs

2. $x < 5$: Then AC will always be a point of MCSTs and CE will never be a part.

   Left→ $^2C_1$ Right → 1 ⇒ Total $^2C_1 * 1 = 2$ MCSTs

3. $x = 5$



⇒Right $^2C_1$ Left $^2C_1$ ⇒Total = $^2C_1 \times {}^2C_1$

$= 2 \times 2$

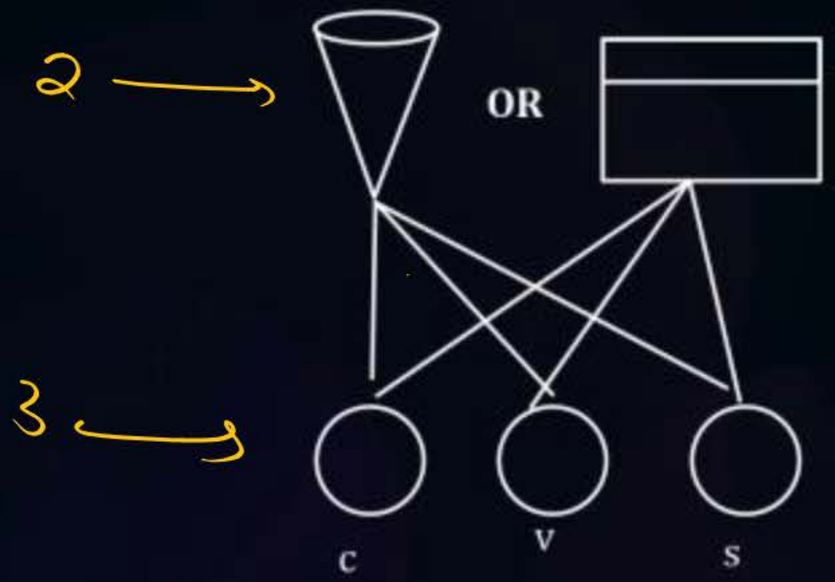
$= 4$ MCSTs



Topic : Greedy Techniques

When to ~~mail~~ mul and add

Ice Cream



How many ways Ice cream can be chosen?

$2 \times 3 = 6$



Topic : Greedy Techniques

PYQ

#Q. G= (V.E) is an undirected simple graph in each edge has a distinct weight, and e is particular edge of G. Which of the following statement about the minimum sapnning tree (MSTs) of G is/are True?

V.Imp

37%

MCST

I. If e is the lighest edge of some cycle in G, then every MST of G includse e.

False

II. If e is the heavist edge of some cycle in G, then every MST of G excluded e.

True

Dijkstra (Kruskal)

A I only

B II only

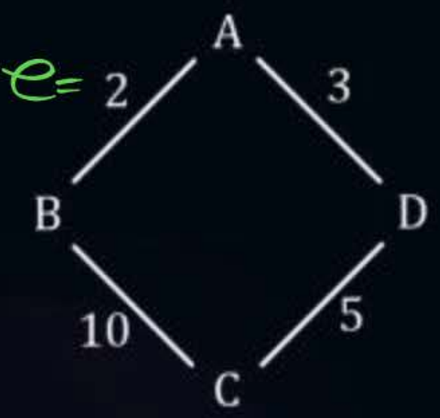
B

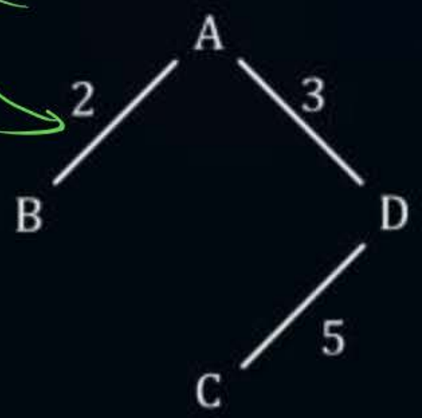
C Both I and II

D Neither I nor II

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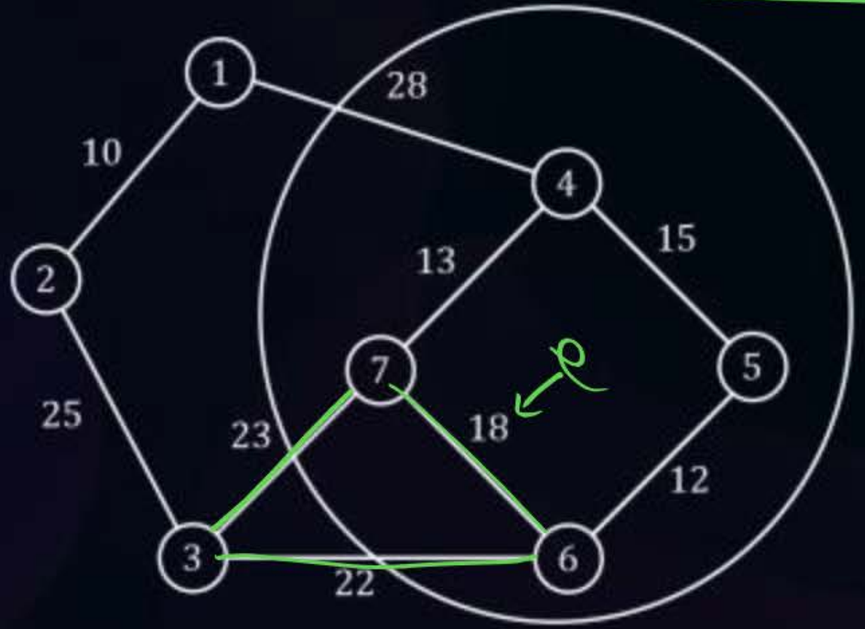
~~Eg.7.~~

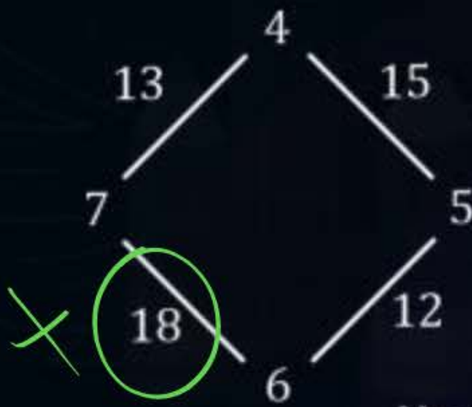




MCST

~~Eg.8.~~





Not be MCST



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62%.

#Q. Let G be a connected undirect weight graph. Consider the following tow statements.

S_1: There exits a minimum weight edge in G which is present in every minimum spanning tree of G. → False

S_2: If every edge in G has distinct weight, then G has a unique minimum spanning tree. → Property

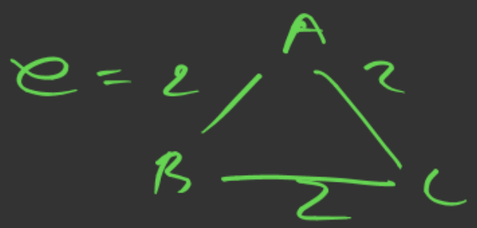
Tow

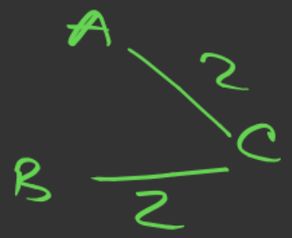
A Both S_1 and S_2 are true

B S_1 is true and S_2 false

C S_1 is false and S_2 is true

D Both S_1 and S_2 are false


e = 2
A
2
B
2
C


A
2
C
B
2



Topic : Greedy Techniques

Kruskal

~~Kensal~~ → End of course (Set + Heap)

Shortest Path Algos

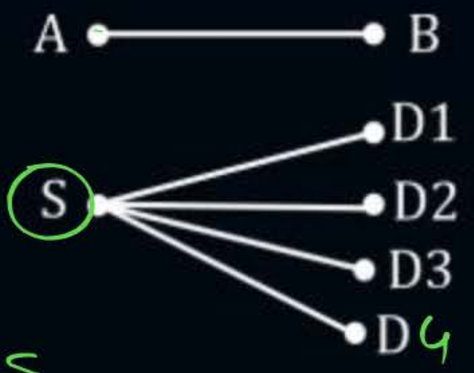


Topic : Greedy Techniques

Shortest Path Algorithms Summary

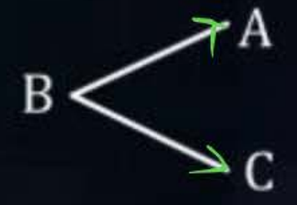
1. Single Pair shortest Path (SPSP):
2. Single source shortest Paths (SSSP)
   - (a) Dijkstra's SSSP Alog→ Greedy
   - (b) Bellman ford→ Dynamic Programming (DP)
3. All pairs shortest Path: (APSP)
   - (a) Floyd war shall algo

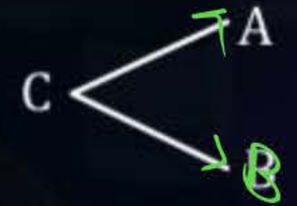
     Dynamic Programming (DP)



A, B, C









Topic : Greedy Techniques

#Q. Dijkstra's Single source shortest Path Algorithm (SSSP)

2. Approaches:-

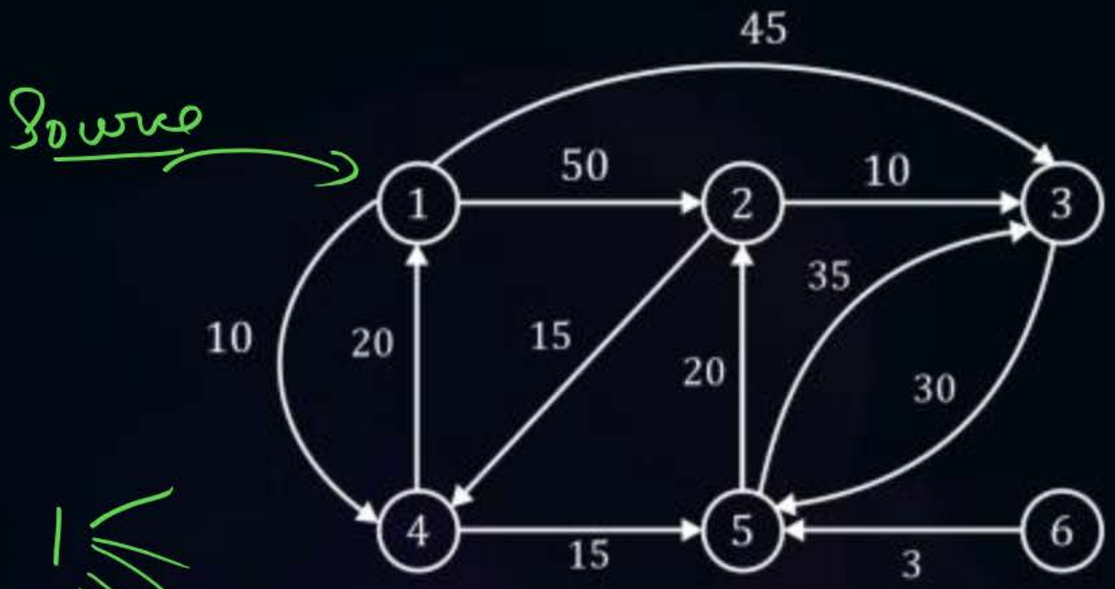
    1. Based on Matrix → given only SSSP cost but not paths
    2. Based spanning Tree→ Gives both costs as well as paths.



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1. Matrix Based

Given:-



Source

Source 6



Relaxation

vertex Set= {}	1 ∞	2 ∞	3 ∞	4 ∞	5 ∞	6 ∞
S1{1}	-	50	45	10	∞	∞
S2{1,4}	-	50	45	10	25	∞
S3:{1,4,5}	-	45	45	10	25	∞
S4{1,4,5,2}	-	45	45	10	25	∞
S5{1,4,5,2,3}	-	45	45	10	25	∞
S6{1,45,2,3,6}	- 0	45	45	10	25	∞

Final ans



Topic : Greedy Techniques

d(x) = distance from source (V0) to a vertex 'x' known so far.

Final answer

$1 \rightarrow 2 : 45$	$1 \rightarrow 4 : 10$
$1 \rightarrow 3 : 45$	$1 \rightarrow 5 : 25$
	$1 \rightarrow 6 : \infty$



Relaxation

Straitly less
If (C2 < C1)
Update d(x) = C2
}

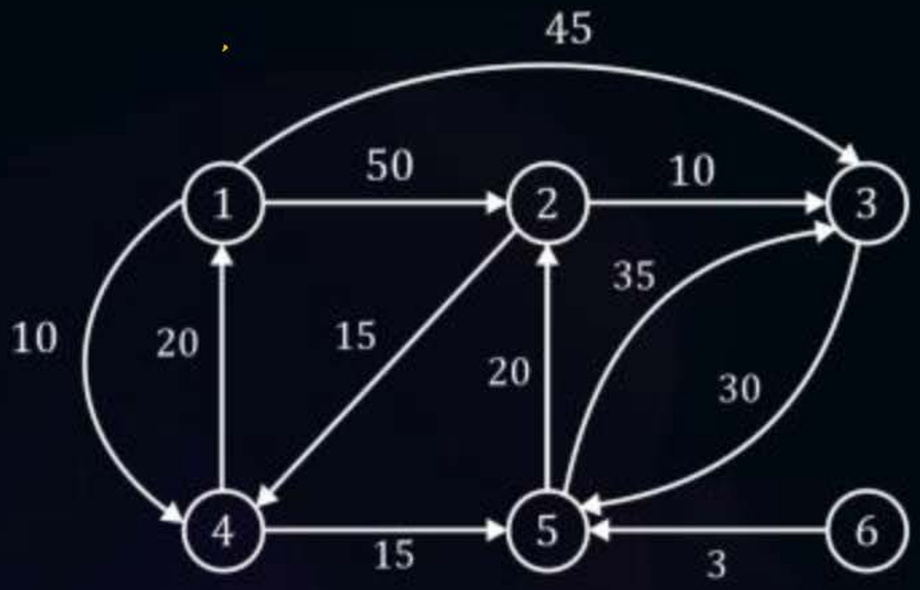


Topic : Greedy Techniques

Practice:- Source

Source = 6

Given:-



vertex Set= {}	1 ∞	2 ∞	3 ∞	4 ∞	5 ∞	6 0
S1{6}	∞	∞	∞	∞	3	-
{6,5}	∞	23	38	∞	3	-
{6,5,2}	∞	23	33	38	3	-
{6,5,2,3}	∞	23	33	38	3	-
{6,5,2,3,4}	58	23	33	38	3	-
{6,5,2,3,4,1}	58	23	33	38	3	-

node

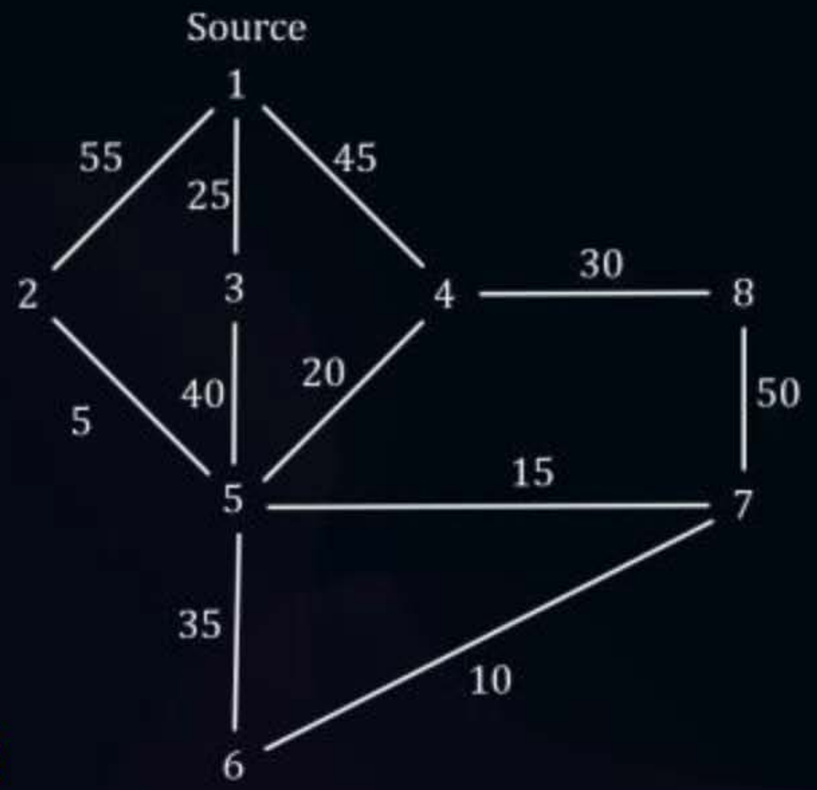
Note :- once a ~~matrix~~ is selected it does not gets furthers relaxed.



Topic : Greedy Techniques

2. Spanning Tree based approach.

Given:



How
↳ Solve & validate using matrix Apps!

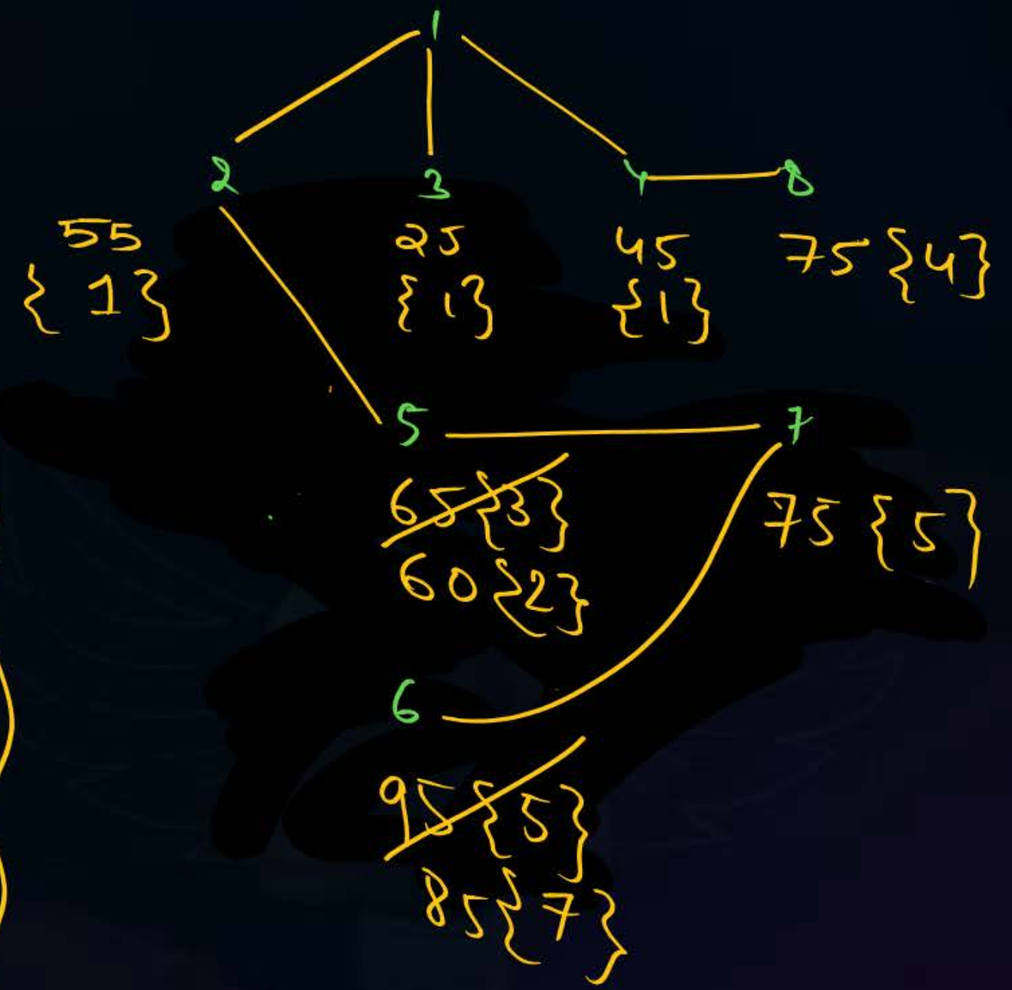
1→ 6?

cost: 85

Path given

By DJ SSSP: [1 → 2 → 5 → 7 →6]

SSSP Spanning Tree





Topic : Greedy Techniques

#Q. Is the spanning tree by above approach of DJ SSSP same as MCST?

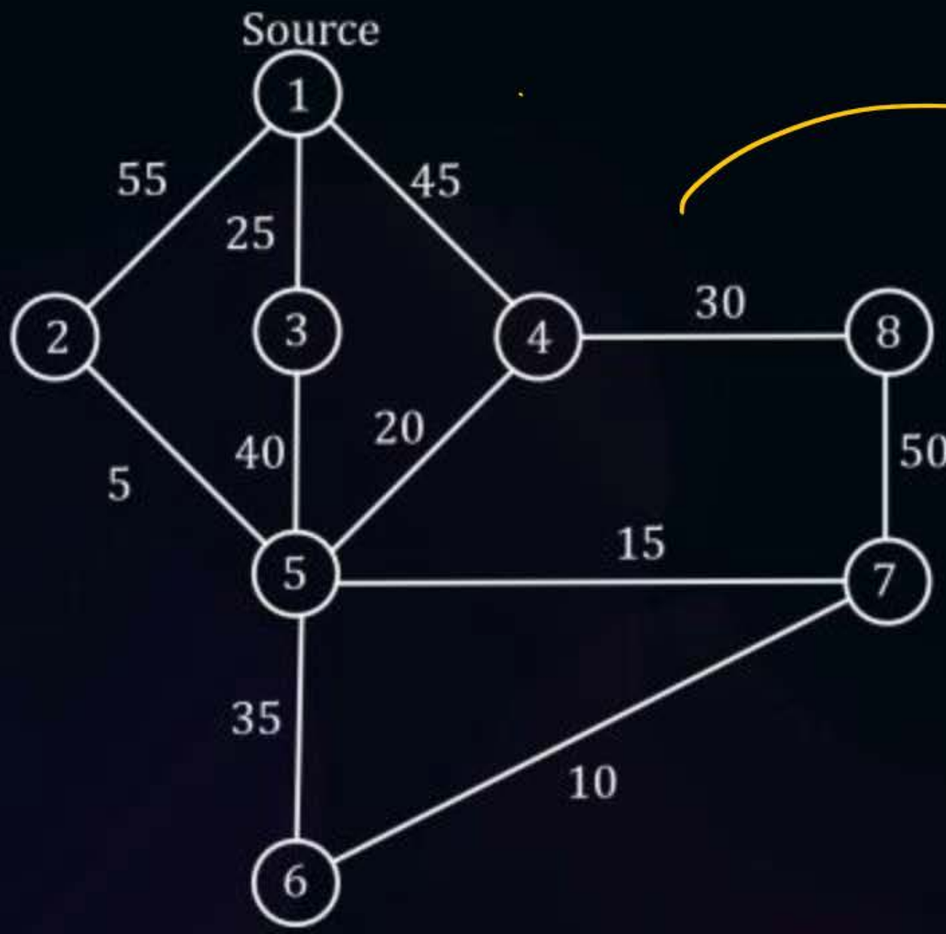
→ No

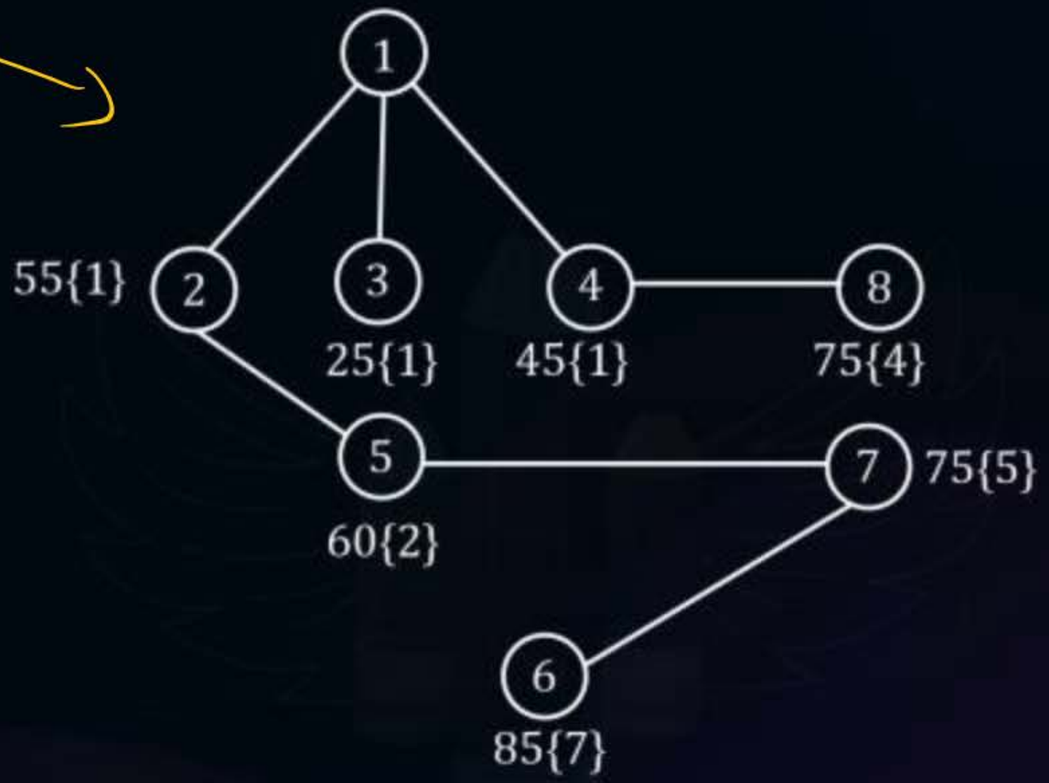
H.W



Topic : Dynamic Programming

DJ SSSP



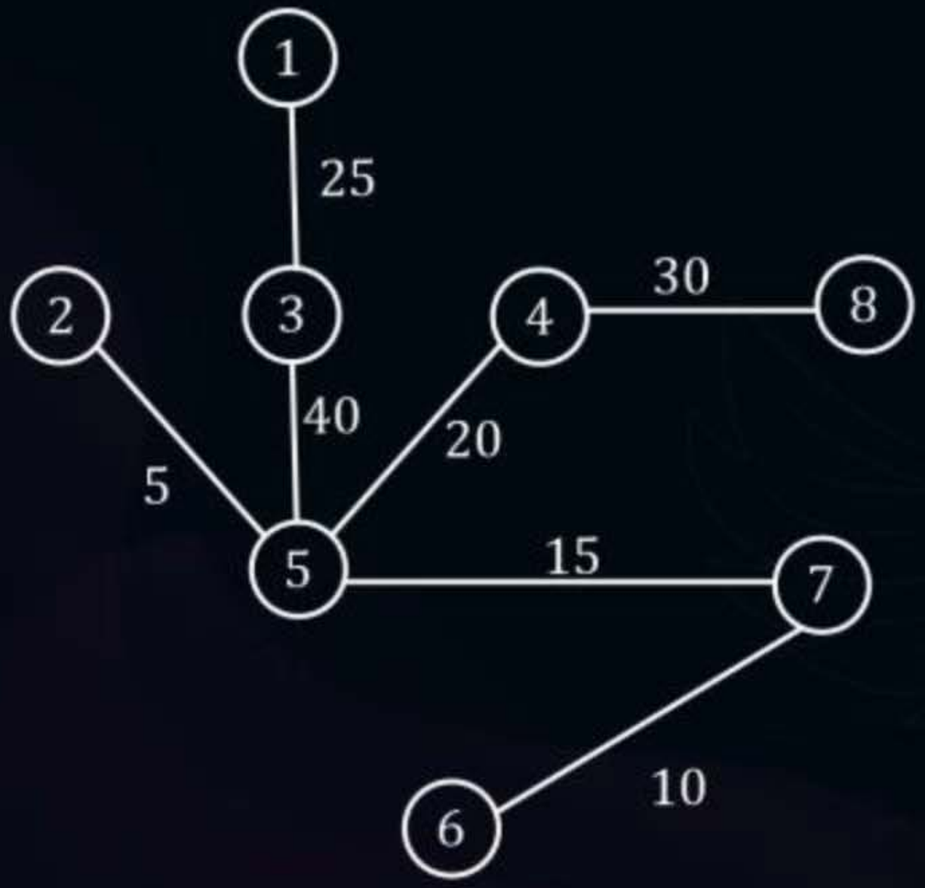




Topic : Dynamic Programming

Spanning Tree:

H.W





Topic : Dynamic Programming

```
1 .   Algorithm ShortestPaths (v, cos dist,n)
2 .   // dist [j], 1 ≤ j ≤ n, is set to the length of the shortest
3 .   // path from vertex v to vertex j in a digraph G with n
4 .   // vertices. dist[v] is set to zero.- G is represented by its
5 .   // cost adjacency matrix cost/.[1: n, 1: n].
6 .         {
7 .         for i := 1 to n do
8 .         {
                  // Initialize S.
9 .               S[i] := false; dist[t] := cost[v, i];
10.    }
11.         S[v] := true; dist[v] := 0.0; // Put v in S.
12    .     for num := 2 to n - 1 do
```



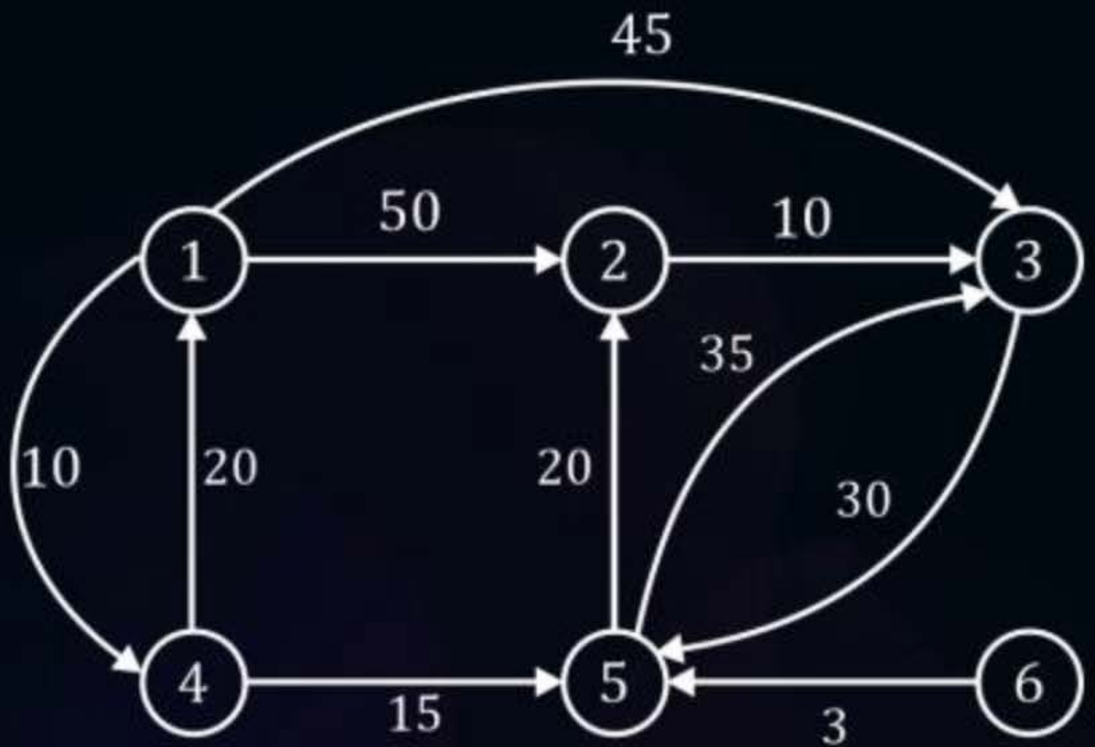
Topic : Dynamic Programming

```
13    {
14          // Determine n - 1 paths from v.
15    Choose u from among those vertices not
16    in S such that dist[u] is minimum;
17    S[u] := true; // Put u in S.
18    for (each w adjacent to u with S[w] = false) do
19    // Update distances.
20    if (dist [w] > dist[u] + cost[u, w])) then
21    dist[w]:= dist[u] + cost[u, w];
22          }
23    }
```



Topic : Dynamic Programming

Source



Set:s	1	2	3	4	5	6
{1}	0	50	45	10	∞	∞
{1,1}	0	-	-	10	-	O(n)
-	-	-	-	-	-	-
-	-	-	-	-	-	-
-	-	-	-	-	-	-



Topic : Dynamic Programming

Time Complexity analysis:-

1. Non-heap based implementation

   TC→ $O(n^2)$

2. Heap based implementation

   TC→ (n+e) logn



Topic : Dynamic Programming

P.Y.Q

#Q. Consider a weighted undirected graph with positive edge weights and let uv be an edge in the ~~edge in the~~ graph. It is known that the shortest path from the sources vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always true?

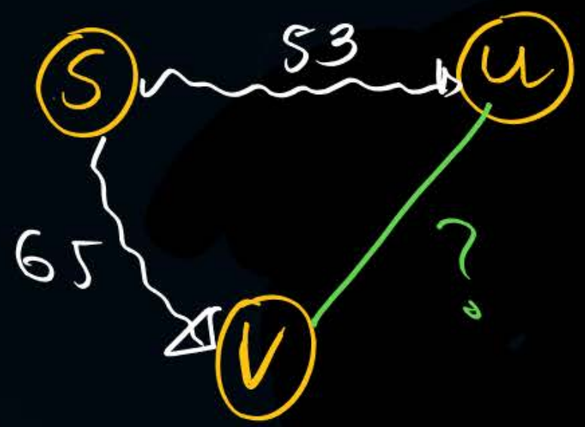
30%

A Weight (u, v) < 12 ✗

B Weight (u, v) ≤ 12 ✗

C Weight (u, v) > 12 → partially correct

D Weight (u, v) ≥ 12 → most correct



given is True, when

$$dist(v) \leq [dist(u) + cost(u,v)]$$

$$65 \leq 53 + x$$

$$\boxed{x \geq 12}$$



Topic : Dynamic Programming

PYQ (Vvgood ques)

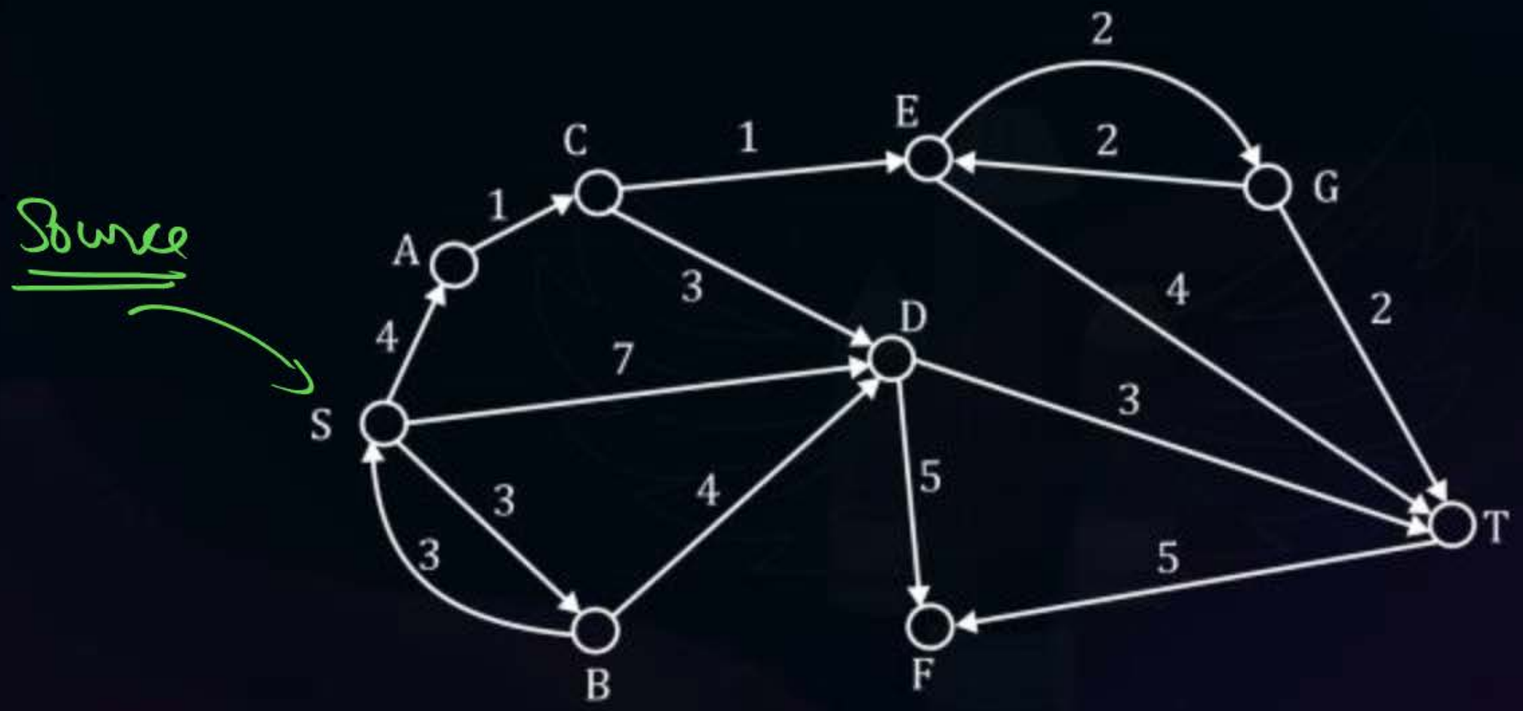
#Q. Applying Dijkstra's Algoirthm over the given Graph, which path is reported for 'S' to 'T';

A SBDT — 22%

B SDT→ → 29%

C SACEGT → 17%

D SACET → 31%



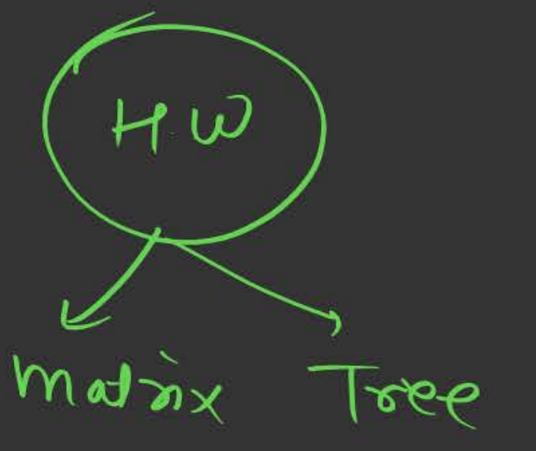
V. V. Common mistake :-

1) SBDT ⇒ 3+4+3 = 10

2) SDT ⇒ 7+3 = 10

3) SACEGT ⇒ 4+1+1+2+2 = 10

4) SACET ⇒ 4+1+1+4 = 10





2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Shortest path

Dijkstra SSSP



THANK - YOU